

PicaCAN: Reverse Engineering Physical Semantics of Signals in CAN Messages Using Physically-Induced Causalities

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Abstract—With the rapid development of Connected and Autonomous Vehicles, In-Vehicle Network attacks have garnered heightened research scrutiny due to vehicles’ increasing connectivities to the external environment. The common characteristic among these attacks is to tamper with targeted powertrain-related signals in the Powertrain Controller Area Network (PT-CAN) and further physically threaten vehicles’ safety. These powertrain-related signals are encoded within CAN messages grounded by the syntax specification, which is proprietary to Original Equipment Manufacturers and publicly unavailable. Thus, to undertake comprehensive security analysis and strategies, reverse engineering PT-CAN to the semantic level is urgently needed. However, the existing methods rely on interactions (injecting challenge signals/actions) with the targeted vehicle, and certain manual efforts are required. To fill this gap, we propose PicaCAN, a novel framework to extract signals from CAN messages and reverse engineer their physical semantics based on physically induced causality. Once access to the CAN traffic, PicaCAN offers the researcher an eye on the vehicle’s powertrain system, decoding binaries flows into powertrain-related signals automatically. We experimentally evaluate PicaCAN on PT-CAN of three automobiles containing two power types. The experimental results show that PicaCAN could successfully extract physical signals representing all targeted semantics (pedals, engine speed, etc.) from two Internal Combustion Engine Vehicles and one Hybrid Electric Vehicle under EV mode.

Index Terms—Controller area network, reverse engineering, vehicular sensory data, physically-induced causalities.

I. INTRODUCTION

CONTROLLER Area Network (CAN) is widely used in in-vehicle communication. Due to its robustness to

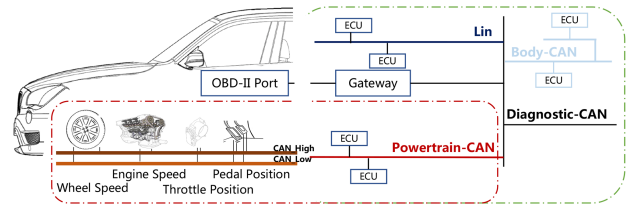


Fig. 1. Common signal interactions on PT-CAN within IVN.

electromagnetic noise, CAN is still the *de facto* communication standard in today’s In-Vehicle Networks (IVN), especially for the most critical powertrain system [1]. As shown in Fig. 1, the devices of the powertrain system (e.g., vehicle engine, accelerator pedal) communicate with each other via the Powertrain CAN (PT-CAN) bus and send/receive real-time physical signals (e.g., engine speed and pedal positions) using CAN messages.

With CAN messages transmitted on the PT-CAN, the powertrain system’s Electronic Control Units (ECUs), such as the Engine Control Unit, can parse the physical signals according to CAN message translation tables, and further generate appropriate control commands to maintain smooth vehicle operations.

With the advancement of Connected and Autonomous Vehicles (CAVs) and Vehicle-to-Everything (V2X) technology [2], automotive networks have become more openly accessible [3], increasing the risk of penetration to PT-CAN. Researchers have identified multiple remote access points in connected vehicles and demonstrated the potential for attackers to exploit these vulnerabilities remotely [4], [5]. Once intruded into the PT-CAN, adversaries can endanger the vehicle by manipulating specific physical signals according to the CAN message translation tables, such as invalidating the brake systems [6] or deceiving the vehicle speed [7].

Although communication protocol J1939 applied in heavy-duty vehicles is open-source [8], the CAN message translation table for the small passenger car is proprietary to the Original Equipment Manufacturers (OEMs) and are not publicly accessible. This drives attackers and security researchers to reverse engineer the physical semantics of signals within the CAN messages of compact passenger vehicles.

The primary objective of CAN reverse engineering is to determine the precise locations of signals within frames, a process known as tokenization, and to decode their semantic meaning

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as well as their format, known as translation. Tokenization is a prerequisite for translation. Only after the signals have been extracted can their physical semantics and formats be rigorously analyzed. Since the work on the former aspect is relatively well-developed, this paper primarily focuses on semantic translation. One of the main approaches for semantic translation is based on Parameter IDs (PIDs) which are codes used to request data from ECUs to provide information on various system parameters and statuses for diagnostics and monitoring. By injecting PIDs via the OBD-II port of the target vehicle, researchers align ECUs' response signals with signals tokenized from CAN trace for semantic translation with high precision [9], [10]. However, since PID is developed for emission-related regulations, PID-based methods can only be applied in fuel vehicles. Other semantic translation approaches are based on executing operations by humans including installing external sensors or conducting action testing within the vehicle and monitoring the CAN trace to capture the triggered signals within CAN traces [11], [12]. In the case that the specifications of PT-CAN are expected to vary depending on the manufacturer, model, and year of the vehicle, executing operation based approach demands significant human and financial resources when facing wide range of vehicles.

In the absence of interactions (such as PIDs, challenge actions, or signals), reverse engineering CAN messages to the semantic level is quite formidable, as the messages within CAN traces are devoid of any labels. Underlying millions of hexadecimal flow through the PT-CAN bus during normal operation of the vehicle, powertrain-related signals have intricate input-output control logic processed by ECUs or complicated hybrid physical dynamics that are difficult to model.

To address the above issues, we design *PicaCAN* to automatically extract signals from CAN messages and reverse engineer their physical semantics, eliminating the need for injecting PIDs or performing challenge signals/actions to the target vehicle. *PicaCAN* is grounded on the fact that the causalities of signals in CAN messages are induced by the dynamics of corresponding powertrain devices, which are marked with physical semantics. By identifying the mappings between CAN signals and the powertrain devices, *PicaCAN* reverse engineers the physical semantics of CAN signals. Specifically, we first capture the dynamics of powertrain devices using a Sequential Event Causality Graph (SECG) according to the mechatronic structure of powertrain systems. Then, taking the CAN messages as inputs, *PicaCAN* automatically extracts the signals based on bit flip rate, and abstracts their causalities using a Timed Event Causality Graph (TECG), in which the nodes represent the extracted signals and the weights of edges quantify the impact latency. *PicaCAN* bridges the above two graphs according to their similarities in structure and edge weights. Steered by the identified mapping, *PicaCAN* reverse engineers the physical semantics of the extracted signals as the properties of powertrain devices.

We have evaluated *PicaCAN* on three commercial vehicles. The experimental results show that *PicaCAN* could successfully extract physical signals representing all targeted semantics (pedals, engine speed, etc.) from two Internal Combustion Engine Vehicles (ICEVs) and one Hybrid Electric Vehicle (HEV)

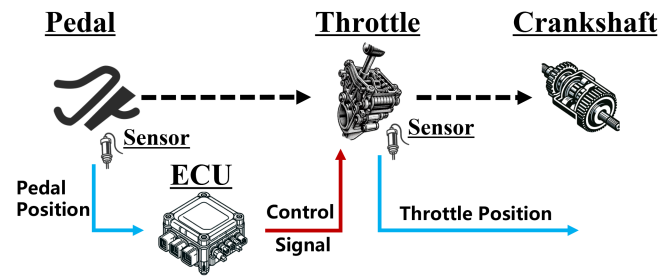


Fig. 2. An exemplary signal flow within a powertrain system.

under EV mode. The contributions of this work are summarized below.

- We propose *PicaCAN*, a novel framework to reverse engineer the physical semantics of signals in CAN messages with limited human effort.
- *PicaCAN* is built on the physically induced causalities — i.e., the dynamics of powertrain devices decide the causalities of signals in CAN messages. These physically induced causalities widely exist in commercial vehicles and make *PicaCAN* pervasively deployable.
- The empirical experiments on three commercial vehicles, including electric vehicles and fuel vehicles, have corroborated the effectiveness of *PicaCAN*.

The remainder of this paper is organized as follows. Section II provides basic about vehicle powertrain systems and communication via PT-CAN. In Section III, intuition towards physically induced causalities is delivered, while Section IV elaborates on SECG generation. Section V details the design of *PicaCAN*. Section VI evaluates *PicaCAN* on three real vehicles followed by discussions given in Section VII. Section VIII offers related research in CAN reverse engineering followed by conclusions in Section IX.

II. PRELIMINARIES

Here, we introduce the powertrain electronic systems and PT-CAN in detail.

A. Electronic Systems of Powertrain

With the development of electronic control systems, many mechanical systems in automobiles, including the most critical powertrain systems, have been replaced by mechatronic systems. Each mechatronic system consists of (i) *sensors* to perceive the states of automobiles, (ii) *Electronic Control Units (ECUs)* to generate control commands based on the sensor readings, and (iii) *actuators* to perform the corresponding control commands. We demonstrate how the three components mentioned above are coupled and function together through the instance of an electronic throttle control system [13]. As shown in Fig. 2, we visualize the power transmission in powertrain flow from the driver's accelerator pedal position to the crankshaft of the internal combustion engine. Taking sensor readings of the accelerator pedal position as inputs, the corresponding ECU generates precise control commands to the throttle motor. The motor then

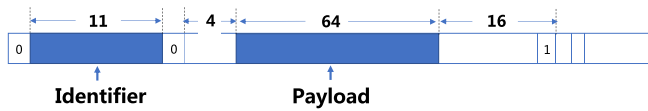


Fig. 3. The specification of standard CAN protocol.

regulates the opening degree of the throttle following the pre-defined control rules [14]. The larger the throttle valve opening, the greater the amount of air entering and the faster up-and-down motion of the pistons. Finally, the power is transmitted out with the crankshaft to drive the vehicle forward.

B. Powertrain-Can

The devices, e.g., sensors, ECUs, and actuators, mentioned above in the powertrain system communicate with each other typically via the PT-CAN network, a high-speed CAN network supporting the communication rate of 1 Mb/s (or 500 kBit/s). Note that PT-CAN has been widely implemented in most commercial automobiles (such as BMW, Benz, and GE) because of its fault tolerance to electrical noise [15].

PT-CAN Communication: Messages in the PT-CAN network are transmitted according to the CAN standard, which specifies the devices thereof communicate in a mode of serial, multimaster, and multicast [16]. Any device connected to the PT-CAN network can broadcast and receive messages following the specification of CAN protocol, as depicted in Fig. 3. Specifically, the physical signals (i.e., sensor readings or control commands) are broadcast to the PT-CAN network by nominating their priorities in the *Identifier* field and encoding them into the *Payload* field. At the same time, the fields of *Data Length*, *CRC*, *ACK*, etc., are also attached.

Note that a singular signal may be initiated to the PT-CAN network using messages with different identifiers (CAN IDs) for different tasks. For example, the *engine_speed* can be sent to Transmission Control Unit (TCU) for gear selection and the dashboard for displaying, using a lower and a higher CAN ID, respectively. The messages with the lower CAN ID have a higher priority and can prevail in the arbitration process if concurrent communication occurs. Also, one CAN message may contain multiple physical signals to save communication resources. As shown in Table I, the message with CAN ID of *0xC4* includes the signals of *Steering_Position* and *Steering_Speed* simultaneously.

CAN Message Encoding/Decoding: Given the above communication patterns, the devices responsible for sending or receiving physical signals must be aware of the corresponding CAN IDs, along with their respective positions within the payload. As shown in Table I, automotive suppliers usually specify such knowledge using a message translation table, named Database CAN (DBC) file, which is private to Original Equipment Manufacturers (OEMs) and not publicly accessible. According to the DBC file, devices can correctly encode and decode the transmitted signals from the CAN bus. As demonstrated in Table I, to provide auxiliary information for the decision-making

TABLE I
BASIC INFORMATION WITHIN THE DATABASE CAN FILE FOR THE
AUTOMOBILE BMW_E9X_E8X1 [17]

CAN ID	Signal Name	Position (bit)	Endianness
0xC4	Steering_Position	0 - 16	Intel
	Steering_Speed	24 - 40	Intel
0xAA	Checksum	0 - 7	Intel
	Counter	8 - 11	Intel
	Accelerator_Pedal	16 - 31	Intel
	Engine_Speed	32 - 47	Intel
	Throttle_Pressed	50	Intel
	Cruise_Pedal_Active	54	Motorola
0xD1	Cruise_Pedal_Inactive	55	Motorola
	YawRate	0 - 16	Intel
	PitchRate	32 - 44	Intel
	Counter	52 - 55	Intel
	CRC	56 - 63	Intel

of TCU, Engine Control Module (ECM) packages the engine speed using 16 bits within the message of ID number *0xAA* and broadcast it via PT-CAN. To obtain the engine speed signal, TCU selects messages with the ID number of *0xAA* on the bus, further analyzing bits positioning from 32 – 47 to determine the optimal timing of gear shifting. In addition to physical signals, there are four other types of signals in the transmitted messages, i.e., *Status*, *Counters*, *Checksum*, and *Constant*, to ensure the bus operates normally [11], [18], [19].

III. PHYSICALLY INDUCED CAUSALITY

Here, we define physically induced causality as Granger causality which follows two principles [20]: 1) the cause occurs before the effect; 2) the cause helps to forecast the effect after other data has been first used. The future values of the effect can be better predicted using the past values of both cause & effect, rather than using only the effect's own past values. We provide three observations to show physically induced causalities within a vehicle's powertrain system.

Observation-I: There are common physically induced causalities between physical signals within the vehicle's powertrain:

As shown in Fig. 2, every time the driver presses the accelerator pedal, ECM receives an accelerator position signal to calculate the opening of the throttle, which would react by throttle electric motor. As the throttle valve increases in size, the amount of air entering the cylinder also increases. With the operation of fuel injection and spark plug ignition, the faster piston movement generates higher power output, which allows for higher engine rotation and finally higher wheel rotation. Note that all executions have a specific order and there must exist causalities behind this system. This causality, induced directly by physical structure, is present in all gasoline-powered and diesel-powered vehicles. For electric-powered or hydrogen-powered vehicles, the chain reaction only needs to be modified by replacing the engine with an electric motor [21].

Physically induced correlation, offering observable correlations in cyberspace, has been widely utilized by researchers

to characterize the relationships between signals within vehicles [1], [12], [22]. Merely indicating statistical relationships between signals, correlation fails to offer directional causal information that how changes in one signal correspond to changes in another. Therefore, the existence of the correlation between two variables is not sufficient to complete reverse engineering work.

Physically induced causality, taking a step forward from correlation, distinguishes cause and effect between pairs of physical signals originating from physical system dynamics. For instance, the input to the ECU is processed through a pre-written program, resulting in an output. If we rely solely on correlation, it is impossible to distinguish between input and output because correlation lacks inherent directionality. In contrast, causality establishes a directional relationship, where the input affects the output, clearly illustrating a cause-and-effect connection in which one acts as the cause and the other as the effect.

Observation-II: The impact of one signal on another requires a certain time, and this impact delay varies for different signals located in different positions of the same loop.

When the driver presses the accelerator pedal, the accelerator pedal position signal carries the driver's requirement on PT-CAN. Consequently, ECM calculates the desired throttle opening degree and sends control signals to the throttle motor, enlarging throttle position signals [13]. The larger degree of the throttle, after undergoing a series of processes in the internal combustion engine, causes higher engine speed. In the sequence from the accelerator pedal position to the throttle position to the engine speed, the delay from the accelerator to the throttle position is larger than the delay from the accelerator to the engine speed.

Observation-III: The impact delay between the two logically nearest physical signals accumulates in magnitude of dozens or hundreds of milliseconds.

For any two logically nearest physical signals, each process to facilitate the flow from one signal to another, including calculation and message transmission, causes time delay [23], which can accumulate to dozens or even hundreds of milliseconds. The minimum sampling period for commonly used normal CAN collecting devices is 10ms, and thus it is possible to capture such delays.

The interaction between physical signals within PT-CAN satisfies two principles in Granger causality as expressed in the *observation*. In particular, the CAN protocol assigns timestamps to each data frame, preserving the sequence of powertrain-related events to ensure the operation of the vehicle. Thus, cause would occur before the effect in the CAN trace. For two physical signals denoted by f_a and f_b , we design the following two *criteria* to detect physically induced causality based on the extracted time delay.

Criterion-I: If the observed impact from physical signal f_a to physical signal f_b decreases with the growth of time delay, then there is no causality from physical signal f_a to f_b .

Since the sampling interval can be sufficiently small, i.e., be a minimum of 10ms, it is reasonable to assume that causality is not immediate. Therefore, when the causality monotonically decreases as the time delay increases, we believe that there is no causal relationship between f_a to f_b .

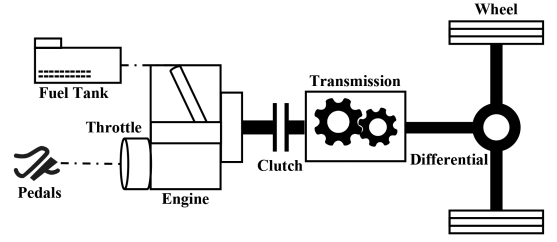


Fig. 4. Schematic of a powertrain system for ICEV [24].

Criterion-II: If the impact delay from physical signal f_a to physical signal f_b is smaller than that from f_b to f_a , then we take physical signal f_a as the cause and physical signal f_b as the effect.

As vehicle systems contain complex negative feedback as well as human interactive behaviors (such as releasing the pedal when the speed is high), the approach to establishing a threshold to determine direction is not universal and applicable. To solve this issue, we introduce an impact time delay between any pairs of physical signals as the primary measure to characterize the physically induced causality.

IV. SECG GENERATION

In this section, we offer an introduction to converting vehicle dynamics of different energy types into SECG, which characterizes the time precedence of changes for powertrain-related signals when the driver activates the pedal positions. The purpose of the SECG is to embed prior knowledge of triggered events within the vehicle powertrain systems, particularly emphasizing the dependencies in their temporal order. We demonstrate that the SECG is easily accessible from the existing open-source materials elaborating on vehicle powertrain systems, which have commonalities in vehicles of the same energy type. To ensure the feasibility of subsequent graph matching, we use the same criteria listed in Section III to generate the SECG as a directed weighted graph, where the edge direction is determined by whether it is the input or output of the same ECU as well as the precedence relationship in a mechanically coupled structure and the edge feature is defined by the sequence of influences and the positive or negative correlation between signals.

A. ICEV's SECG

Based on the schematic of the powertrain system for ICEVs depicted in Fig. 4, we construct the corresponding SECG in Fig. 5. For the temporal relation between the brake pedal position and its receiving point, we are unsure if the brake-by-wire system is adopted by the target vehicle. We just keep the direction of the edges on the SECG and the sign of the edge weight (negative correlation). For SECG $G_q = (V_q, A_q, E_q)$, we have:

$$V_q = \{1: \text{AcceleratorPedalPosition}; 2: \text{BrakePedalPosition}; \\ 3: \text{ThrottlePositionSignal}; 4: \text{EngineSpeedSignal}; \\ 5: \text{VehicleSpeedSignal}\}$$

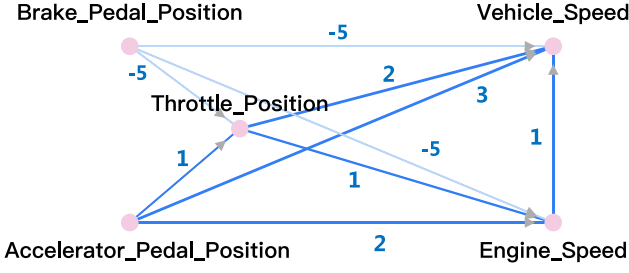


Fig. 5. SECG constructed for ICEV.

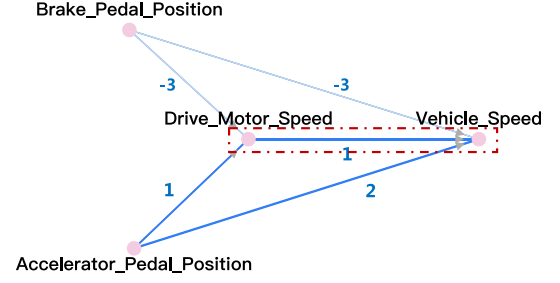


Fig. 7. SECG constructed for HEV under EV mode.

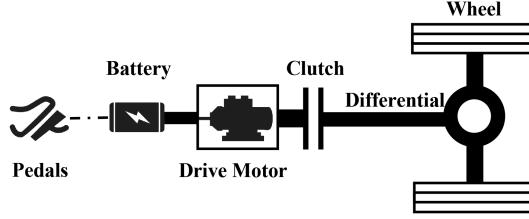


Fig. 6. Schematic of a powertrain system for HEV under EV mode [25].

$$\mathbf{A}_q = \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \mathbf{E}_q = \begin{bmatrix} 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & -5 & -5 & -5 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

where $\mathbf{A}_q$ and $\mathbf{E}_q$ follow the rules introduced in Section III. Each row/column is mapped to specific elements in set $\mathbf{V}_q$ ordered according to natural numbers. Entries $\mathbf{A}_q(i, j)$ and $\mathbf{E}_q(i, j)$ denote for causal relation between i th and j th nodes in set $\mathbf{V}_q$. The constructed SECG is prepared for the ICEVs' reverse engineering in Section VI.

B. HEV Under EV Mode's SECG

As shown in Fig. 6, the powertrain of HEV under EV mode is fundamentally different from ICEVs. Specifically, receiving the pedal position signal, the power battery regulates the magnitude of the three-phase electric current output as needed, thereby controlling the rotational speed of the drive motor. Due to the regenerative braking mechanism, the motor is driven by the battery when the accelerator pedal is greater than zero. When the pedal is equal to zero and the motor speed exceeds zero, the motor generates electricity for the battery. In addition, BYD Han is equipped with a single-speed transmission, resulting in a strong correlation between drive motor speed and vehicle speed. Considering the above characteristics, we generate the SECG for HEV under EV mode displayed in Fig. 7. We add a red box between the drive motor speed and the vehicle speed to visualize the constraint that is required within the same semantic set. Besides, we exclude the three-phase alternating current because its underlying causalities, along with others, are variable.

V. SYSTEM DESIGN

In this section, we detail the design, named *PicaCAN*, which can reverse engineer semantic meanings of physical signals on PT-CAN based on physically induced causality. *PicaCAN* includes six steps, which is illustrated in Fig. 8. The input of *PicaCAN* is the CAN trace obtained from a gateway or OBD-II port and the target vehicle's power type. The vehicle's power type guides the selection of the SECG mentioned in Section IV. The normal CAN trace, containing billions of binary bit streams, is processed in ① Physical Signal Extraction to extract physical signals by detecting monotonically decreasing bit-flip rates [26].

From ② Pre-Processing to ④ TEGC Construction, a direct weighted causal graph is generated based on the input normal CAN trace where ② Pre-Processing deals with the determination of nodes of the TEGC: physical signals are clustered into semantic sets by the criteria that physical signals with the same semantics have a strong correlation. As for edges between pairs of semantic sets, the direction indicates cause/effect in physically induced causalities and the weight denotes by time delay between the pairs. We design *RatioDSTE* in ③ RatioDSTE Calculation to catch impact delay between pairs of signals which is the key indicator to determine the direction and weight of the edges based on Criteria-I and II. Finally, in ⑤ Subgraph Matching, we TEGC and SECG with similarities based on structural and weight similarities. The two graphs are in the same form of directed weighted and follow the same criteria. Comparing the topology and edge characteristics of $\mathbf{G}$ and $\mathbf{G}_q$, we complete node mapping between two graphs to determine the corresponding node pair representing the same physical semantics, eventually acquiring in-vehicle data related to the powertrain systems.

A. Physical Signal Extraction

Here, the physical signal is extracted using its distinctive variation characteristics, i.e., lower-order bits exhibit higher bit-flip rates within the CAN bus encoding environment [26]. Without loss of generality, we assume that manufacturers use a 16-bit encoding scheme for vehicle speed. The changes in vehicle speed within a 10ms sampling period are constrained. Given that vehicle speed does not undergo abrupt changes at adjacent sampling times, the lower position bits change first, followed by the higher bits when the vehicle speed increases or

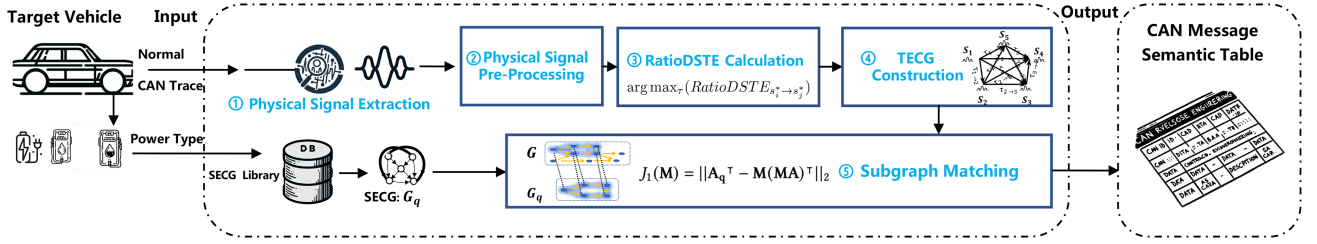


Fig. 8. Main work flow of PicCaCAN.

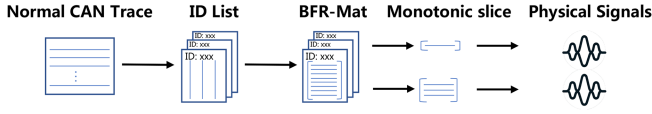


Fig. 9. Main work-flow for Physical Signal Extraction.

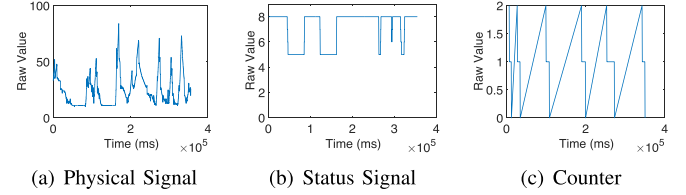


Fig. 10. Raw Values of different categorized signals.

decreases. The bit flip rate, determined by dividing the number of bit changes in the payload by the total occurrences, serves as an effective measure for assessing the rate of bit changes.

As depicted in Fig. 9, clustering is initially conducted based on the ID to generate a CAN trace associated with the same ID. Subsequently, a bit-flip rate matrix (BFR-Mat) is generated for each ID's Payload field, where the 64-bit flip rate is sequentially written into the $R^{8 \times 8}$ matrix row by row. For each row in BFR-Mat, we extract the sequence with a monotonic value decreasing from low to high within a single byte. For all sequences extracted from the same BFR-Mat, it is imperative to ascertain whether the extracted signal is single-byte encoded or multi-byte encoded. In the case of multi-byte encoding, an additional step involves identifying whether it follows Motorola or Intel encoding, ensuring the accurate extraction of a signal. Sequences are considered to be single-byte encoded and can be directly extracted when either of the following situations occur:

- The sequence is not encoded from bit 0;
- The sequence is encoded from bit 0 within the byte, but the sequence of neighboring bytes does not satisfy the full-byte encoding or monotonically requirements.

If the byte is encoded with full bits and its neighboring bytes are encoded from bit 0, then the consequence is multi-byte encoded. The basis is that the bit-flip rate encoded for the lowest bit of its adjacent byte is less than the highest bit-flip rate of the byte itself. To further identify whether it follows Motorola or Intel encoding, the rules are given for each full-encoded byte below.

- If the bit-flip rate encoded for the lowest bit of the last byte is less than the highest bit-flip rate of the byte itself, the sequence is **Motorola encoded**;
- If the bit-flip rate encoded for the lowest bit of the next byte is less than the highest bit-flip rate of the byte itself, the sequence is **Intel encoded**;
- If the last and next bytes are encoded with full bytes, then merge them with the byte itself and continue to search for more bytes by repeating the steps above.

The single-byte and multi-byte encoded sequences are then converted to decimal according to their encoding format.

According to the encoding feature released in [11], [26], we remove all other types of CAN messages including status/counter illustrated in Fig. 10(b) and (c), respectively. The output of ① Physical Signal Extraction is n physical signals ($f_1, f_2, \dots, f_n$) illustrated in Fig. 10(a). Each physical signal represents a temporal signal that is a sequence of data points (signal value) with a sequence of associated timestamps.

B. TEGC Generation

In this subsection, we introduce how to formally define and apply physically-induced casualties to construct a directed weighted causal graph from multiple physical signals extracted by ① Physical Signal Extraction. Our objective is to visualize the interaction pattern between various physical signals when the vehicle is under normal operation. The causal discovery techniques have been widely used in detecting coupling between dynamic systems [27], [28]. Considering the message characteristics of the CAN bus, we propose Double Symbolic Transfer Entropy (DSTE) modified from Symbolic Transfer Entropy (STE) to decouple the relationships among various components within the automotive system from a purely data-driven perspective.

Physical Signal Pre-Processing: First, we tackle the challenges of determining nodes (semantic sets) for the TEGC. Our original intention is to represent all the extracted physical signals with as few semantic sets as possible. We observe that although there are hundreds of physical signals active on the CAN bus [29], the types of the semantics of these physical signals are usually far fewer as there are only approximately ten distinct types of in-vehicle sensors. Commonly, multiple physical signals transmitted on the CAN bus convey the same semantics [18]. The most direct reason is that the vehicles contain many sensors measuring the same semantics, e.g., four-wheel speed sensors, which in different positions use different encoding specifications. Also, ECUs would modify physical

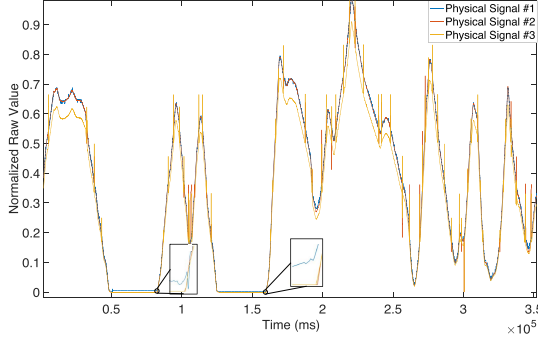


Fig. 11. A visualization of pivot signal at the earliest time scale within the semantic set.

signals received from different sensors and package them to send to other ECUs, causing the fact that various physical signals released at different timestamps have the same physical semantics. As physical signals with the same semantics must have a strong correlation, ③ Physical Signal pre-processing is developed to yield a small number of semantic sets covering all physical signals.

Taking time delays between any pair of physical signals with the same semantics as input, we apply lightweight *Time-Delayed Pearson Correlation Coefficient* to cluster physical signals having identical semantics together into one semantic set. Although the factors or offsets defined by DBC files for the same semantic signals may be different, their *Pearson Correlation Coefficient* is close to one because of the strict linear relationship within signals having the same semantics. We use *time-delay* to find the earliest signals within each semantic set and take it as the pivot signal.

Notice that due to the internal mechanism of CAN, different message IDs (identifiers) are transmitted on the bus at different frequencies, resulting in inconsistent sampling frequencies for signals within different IDs. Therefore, the sequence of timestamps for different physical signals may vary. Given a pair of physical signals as X and Y extracted within set $(f_1, f_2, \dots, f_n)$, we first adopt linear interpolation to reconstruct two signals in their overlap time. We denote the pair of physical signals in the same sequence of timestamps as $\bar{X}$ and $\bar{Y}$. Pearson Correlation calculation with time delays τ can be written as:

$$\rho_{\bar{X}, \bar{Y}_\tau} = \frac{\text{cov}(\bar{X}, \bar{Y}_\tau)}{\sigma_{\bar{X}} \sigma_{\bar{Y}_\tau}} = \frac{E[(\bar{X} - \mu_{\bar{X}})(\bar{Y}_\tau - \mu_{\bar{Y}_\tau})]}{\sigma_{\bar{X}} \sigma_{\bar{Y}_\tau}} \quad (1)$$

where $\bar{Y}_\tau$ is lagging τ time unit to $\bar{Y}$. For each pair of physical signals, we iteratively compute $\rho_{\bar{X}, \bar{Y}_\tau}$ with $\tau \in [-100, 100]$ time unit and select the maximum value $(\rho_{\bar{X}, \bar{Y}_\tau})_{\max}$ and its corresponding time delay τ . We group physical signals with $\rho_{\bar{X}, \bar{Y}_\tau} \geq \theta_0$ as a semantic set and choose the physical signal with the smallest time delay as the pivot signal, where $\theta_0 < 1$ and we set θ_0 near 1. As the example illustrated in Fig. 11, three physical signals are clustered together due to a high $\rho_{\bar{X}, \bar{Y}_\tau}$, and from the magnified details, it can be observed that there is a certain time delay between the signals and the first physical signal is chosen as the pivot signal for this semantic set. For most of cases, one

semantic set would only have one pivot signal. However, there are some special cases in two signals with different physical semantics that appear to have $\rho_{\bar{X}, \bar{Y}_\tau}$ close to 1 in the vehicle, which would be depicted in the SECG. To address this, we need to further split the semantic set according to time delay τ .

The output of ② Physical Signal Pre-Processing are multiple semantic sets $\{v_1, v_2, \dots, v_N\}$ with their corresponding pivot signal $\{s_1, s_2, \dots, s_N\}$, where N denotes number of semantics sets.

By clustering identical semantic physical signals, it effectively reduces the number of nodes of the graph from n to N , greatly reduces the computational complexity, and avoids redundant calculations of DSTE with Time Delay in ③ RatioDSTE Calculation. For each semantic set, selecting the signal at the earliest time scale is crucial to prevent causal reversal situations in subsequent causal discovery stages due to time lags.

Double Symbolic Transfer Entropy with Time Delay Calculation: With determined pivot signals representing the node (semantic sets) of the TEGC, we focus on measuring causal relationships between these nodes here. These nodes represent the signals from a certain sensor in the vehicle's powertrain, e.g., speed, engine speed, etc.

Facing the challenge that these physical signals are non-stationary and the relationship between pairs of physical signals is non-linear, Symbolic Transfer Entropy (STE) based methods turn out to be the first choice.

STE-based methods generate ranking vectors to reconstruct a stationary time series based on the original one [30], [31]. However, the existing STE methods do not distinguish cases in that there are constant intervals within the original signal [30]. For instance, the ranking vectors for the original value array (10,10,10) and (8,9,10) are all (1,2,3). While the former case is very common in the CAN bus communication mechanism, using the existing STE methods can make the probability estimation extremely inaccurate, thereby leading to unreliable results in TE calculation. Therefore, we proposed a modified algorithm, named Double Symbolic Transfer Entropy, to mitigate problems mentioned in TE calculation [32]. Under DSTE, the ranking vectors for (10,10,10) would be [1, 2, 3; 3, 2, 1] while for (8,9,10) is [1, 2, 3; 1, 2, 3]. The details of DSTE are given as follows.

Given each pair of pivot signals in set $\{s_1, s_2, \dots, s_N\}$, same as processed before, aligning their timestamps is the first step. For convenience, we denote the pair of pivot signals after aligning as X^* and Y^* . We give their embedding vectors in the form of $\mathbf{x}_t^* = [x_t^*, x_{t-t'}^*, \dots, x_{t-(k-1)t'}^*]$ at time instant t and $\mathbf{y}_t^* = [y_t^*, y_{t-t'}^*, \dots, y_{t-(k-1)t'}^*]$, where k denotes for the embedding dimension and t' is the scaling time interval in embedding vectors. The values of k and t' would affect the probability estimation of variables as well as the magnitude of Entropy. The selection of k and t' is based on pivot signals collected in ③ Physical Signal Pre-Processing and we determine the parameters by examining the entropy values among all pivot signals. In Fig. 12, the entropy increases continuously as the parameter increases and eventually stabilizes. We then set $k = 6$ and $t' = 20$ such that the entropy of most pivot signals starts to stabilize.

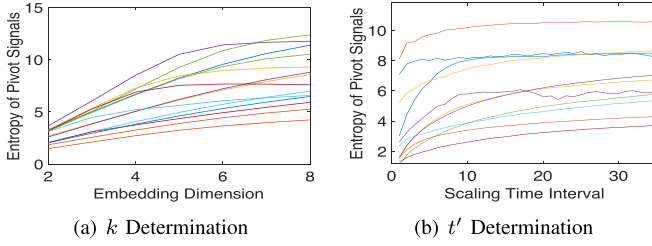


Fig. 12. Determination of embedding dimension and scaling time interval in TEGC Generation.

Given the embedding vectors of pivot signal X^* , we present its sequence arranged in ascending order in the following form $\{x_{t-(r_{t,1}-1)t'}^* < x_{t-(r_{t,2}-1)t'}^* < \dots < x_{t-(r_{t,m}-1)t'}^* = x_{t-(r_{t,m+1}-1)t'}^* = \dots = x_{t-(r_{t,m+l-1})t'}^* < \dots < x_{t-(r_{t,k}-1)t'}^*\}$ at arbitrary time point t where $r \in [1, k]$ and $(m \sim m+l)$ indicates possible equality situations and the order is associated with index k [30]. We describe $\mathbf{x}_t^*$ with two symbolic sequences defined below

$$\begin{aligned} \vec{x}_t^* &= [r_{t,1}, \dots, r_{t,m-1}, r_{t,m}, \dots, r_{t,m+l}, r_{t,m+l+1}, \dots, r_{t,k}] \\ \overleftarrow{x}_t^* &= [r_{t,1}, \dots, r_{t,m-1}, r_{t,m+l}, \dots, r_{t,m}, r_{t,m+l+1}, \dots, r_{t,k}] \end{aligned} \quad (2)$$

$\vec{x}_t^*$ and $\overleftarrow{x}_t^*$ remain consistent in all other cases except when there are equal slices in $\mathbf{x}_t^*$, emphasizing the characteristic description of equality scenarios which is common in our original signals. For ease of representation, we use symbolic vector $\mathbf{S}_{x,t}^*$ to denote $[\vec{x}_t^* \overleftarrow{x}_t^*]$.

Given symbolic vector $\mathbf{S}_{y,t}^*$ and $\mathbf{S}_{x,t}^*$, we define Double Symbolic Transfer Entropy with time delay τ as

$$\begin{aligned} DSTE_{X^* \rightarrow Y^*}^\tau &= \sum p(\mathbf{S}_{y,t+\tau}^*, \mathbf{S}_{y,t}^*, \mathbf{S}_{x,t}^*) \\ &\quad \times \log \frac{p(\mathbf{S}_{y,t+\tau}^* | \mathbf{S}_{y,t}^*, \mathbf{S}_{x,t}^*)}{p(\mathbf{S}_{y,t+\tau}^* | \mathbf{S}_{y,t}^*)}, \end{aligned} \quad (3)$$

where probabilities $p(\cdot)$ are calculated by frequency histogram statistics based on time series with tens of thousands history of data. Then, we define the Ratio Double Symbolic Transfer Entropy as

$$RatioDSTE_{X^* \rightarrow Y^*}^\tau = \frac{DSTE_{X^* \rightarrow Y^*}^\tau}{H(\mathbf{S}_{y,t+\tau}^* | \mathbf{S}_{y,t}^*)}, \quad (4)$$

where $H(\mathbf{S}_{y,t+\tau}^* | \mathbf{S}_{y,t}^*)$ denotes for conditional entropy [33].

TEGC Construction: Given the derived semantic sets and Ratio Double Symbolic Transfer Entropy, a directed weighted TEGC can then be constructed to represent inner logic between semantic sets, which is defined as:

$$\mathbf{G} = (\mathbf{V}, \mathbf{A}, \mathbf{E}) \quad (5)$$

where $\mathbf{V} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_N\}$ is a set of nodes denoting for the corresponding semantic sets, $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the asymmetric adjacency matrix, matrix $\mathbf{E} \in \mathbb{R}^{N \times N}$ describes edge features between any pair of semantic sets. We randomly assigned numbers from $1 \sim N$ to semantic sets clustered in pre-processing

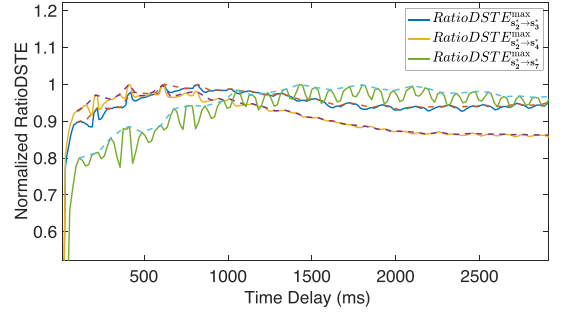


Fig. 13. Determination of edge feature τ_{max} based on the maximum values of $RatioDSTE$.

stage as $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_N\}$, the corresponding pivot signals in each set denote as $\{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_N\}$.

To determine the direction between two nodes, we formulate the criterion-II based on $RatioDSTE$ calculated by (4) of pairwise pivot signal:

$$\begin{aligned} \forall i, j \in \{1, 2, \dots, N\}, i \neq j, \\ \tau_{\mathbf{s}_i^* \rightarrow \mathbf{s}_j^*}^{\max} < \tau_{\mathbf{s}_j^* \rightarrow \mathbf{s}_i^*}^{\max} \Rightarrow \mathbf{A}(i, j) = 1 \end{aligned} \quad (6)$$

where $(\mathbf{s}_1^*, \mathbf{s}_2^*, \dots, \mathbf{s}_N^*)$ are symbolization for pivot signals $(\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_N)$ and $\tau_{\mathbf{s}_i^* \rightarrow \mathbf{s}_j^*}^{\max} = \arg \max_{\tau} (RatioDSTE_{\mathbf{s}_i^* \rightarrow \mathbf{s}_j^*}^\tau)$. The reason we use $RatioDSTE$ instead of $DSTE$ is that the former one could better describe the impact of $\mathbf{s}_i^*$ on $\mathbf{s}_j^*$ as τ varies. When τ increases from zero, it is not just $DSTE_{\mathbf{s}_i^* \rightarrow \mathbf{s}_j^*}$ changes, but also for $H(\mathbf{S}_{y,t+\tau}^* | \mathbf{S}_{y,t}^*)$, which indicates that the magnitude of the influence on signal $\mathbf{s}_j$ by external factors is not constant and is highly necessary to be considered. In this case, we choose $RatioDSTE_{\mathbf{s}_i^* \rightarrow \mathbf{s}_j^*}$ to evaluate when the impact of $\mathbf{s}_i$ on $\mathbf{s}_j$ reaches its maximum. $\tau_{\mathbf{s}_i^* \rightarrow \mathbf{s}_j^*}^{\max}$ can not be obtained if $RatioDSTE_{\mathbf{s}_i^* \rightarrow \mathbf{s}_j^*}$ is monotonically increasing or decreasing with the increase of τ , which reflects that the maximum impact of $\mathbf{s}_i^* \rightarrow \mathbf{s}_j^*$ cannot be captured by a sufficiently wide range of τ .

To determine the edge feature between two nodes with the determined direction, we use the Pearson correlation coefficient to characterize the positive/negative correlation between signals, denoted by positive and negative signs. The specific criterion is followed,

$$\mathbf{E}(i, j) = \begin{cases} \tau_{\max}, & \rho_{\mathbf{s}_i, \mathbf{s}_j} > 0 \text{ and } \mathbf{A}(i, j) = 1 \\ -\tau_{\max}, & \rho_{\mathbf{s}_i, \mathbf{s}_j} < 0 \text{ and } \mathbf{A}(i, j) = 1 \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

where $\rho_{\mathbf{s}_i, \mathbf{s}_j}$ denotes Pearson correlation coefficient between two pivot signals. We offer a visual example of the determination of $\tau_{\max}$ in Fig. 13. We select $\tau_{\max}$ when $RatioDSTE$ reaches its maximum. Based on (6) and (7), we generate a directed weighted graph with exactly one weighted edge between every two nodes.

C. Subgraph Matching

Here, a graph matching between the TEGC generated from normal CAN trace and the selected SECG is provided. Given the SECG as the query graph $\mathbf{G}_q$ and the TEGC generated from data as graph $\mathbf{G}$, we aim to find the sub-graph of $\mathbf{G}$

isomorphic to G_q based on the similarities in structure and edge weights between these two graphs. To achieve this objective, we first adopt a modified enumeration algorithm to find subgraph structure isomorphism [34]. Because the filtered isomorphic graphs may not be unique, we further select the graph that is closest to G_q by comparing their edge weights.

Inspired by [34], we develop a structure matching method, which applies to directed graphs. Two graphs are given in form: $G_q = (V_q, A_q, E_q)$ and $G = (V, A, E)$, with asymmetric adjacent matrix $A_q \in \mathbb{R}^{p \times p}$ and $A \in \mathbb{R}^{N \times N}$, respectively. The optimal correspondence matrix $M \in \mathbb{R}^{p \times N}$ between nodes of two graphs, such that the cost function $J_1(M)$ is minimized:

$$J_1(M) = \|A_q^T - M(AA)^T\|_2 \quad (8)$$

where A_q^T denotes the transpose of matrix A_q and matrix M is constrained to be a one-to-one mapping [35]. Following guidance of matrix M , we could reconstructed edge feature matrix for the selected subgraph $G_s = (V_s, A_s, E_s)$.

Note that the optimal solution for cost function (8) may not be unique, i.e., we can have multiple solutions of M and each M represents an isomorphic sub-graphs in G compared to G_q . Then, we further determine which M best fits based on minimizing difference $J_2(E_s) = \|E_q - E_s\|_2$. However, the edge weight calculated could vary from vehicle to vehicle, also influenced by other factors such as data size, driving condition, etc. Therefore, modification towards matrix E_s is necessary to enhance the versatility of the system. Given the modified edge feature matrix as E_s^* , we assign values in E_s^* as the ranking of this edge among all outgoing edges from its source node based on values in E_s . The cost function convert into $J_2(E_s^*) = \|E_q - E_s^*\|_2$. The output is the exact correspondence M^* . Each 1 in i th row and j th column of M^* indicates that j th node of a TCG G represent the same semantics of i th node of graph G_q , finally extracting in-vehicle data within the normal CAN trace.

VI. EVALUATION

In this section, we thoroughly evaluate Picacan through a two-step procedure. First, we compare the proposed causal discovery method *RatioDSTE* with existing mainstream algorithms to demonstrate its effectiveness in uncovering underlying causal relations among powertrain-related signals. Second, we validate Picacan on both Internal Combustion Engine Vehicle (ICEV) and Hybrid Electric Vehicle (HEV) under EV mode. The normal CAN traces are collected when volunteers drive vehicles.

A. RatioDSTE-Based Causal Discovery Method

As the most critical component of Picacan, *RatioDSTE* is responsible for the causal discovery, extracting the correct pattern of the powertrain-related signals. Thus, the performance of *RatioDSTE* determines the success of reverse engineering.

Here, we perform a comparison between the proposed *RatioDSTE* and the following algorithms for causal discovery in time series data:

- *Temporal Causal Discovery Framework (TCDF)* targets on discovering non-linear causal relation between time series

using deep learning networks with an attention mechanism within dilated depth-wise convolutional network [36].

- *Multivariate Granger Causality (MVGC)* is a representative method of discovering Granger Causality using vector autoregressive model [37].
- *Symbolic Transfer Entropy (STE)* utilizes the technique of symbolization to calculate transfer entropy, overcoming challenges associated with the probability estimation on non-stationary time series [30].

For dataset, we select four powertrain-related signals translated from CAN messages and present the causal discovery results using a summary causal graph as depicted in [38], where nodes denote signals and the edge direction represents the causality. The semantics of the four nodes are provided here: S_1 := Accelerator Paddle Position, S_2 := Engine Speed, S_3 := Throttle Position, S_4 := Vehicle Speed. As illustrated in Fig. 15, only the proposed *RatioDSTE*-based method successfully identifies the correct causal relationships among the four physical signals while *MVGC* and *STE* generates results that are partially correct and *TCDF* fails to return any causal link.

Furthermore, we divide the collected dataset into segments of different temporal lengths from 50% to 100% of the total time length, aiming to test performance of *RatioDSTE*-based method compared with *MVGC* and *STE* under different temporal length using *FI-score*. Fig. 17 shows that the accuracy of the *RatioDSTE*-based method improves as the collection time of CAN logs increases, ultimately reaching 100%. Since *MVGC* relies on the assumption that the time series are stationary [37], its performance on the non-stationary physical signal is limited. The *STE*-based method can address non-stationary time series by symbolizing the data. However, it fails to distinguish between equal and monotonically increasing intervals as mentioned in Section V-B. *RatioDSTE*-based method overcomes this limitation by employing a double symbolic approach, which is illustrated by Fig. 17.

B. Picacan on ICEV

In this subsection, we test Picacan on two ICEVs, i.e., E84 BMW X1 and Renault Koleos shown in Fig. 14(a) and (b), respectively. We collect normal CAN data through a P-CAN device [39] and perform offline analysis on a personal computer. To obtain ground truth, we also acquire diagnostic CAN data from the OBD-II port expressing some common in-vehicle physical semantics, i.e., vehicle speed, engine speed, etc., using OBD-II protocol [40].

Data Collection: For BMW X1, we are unable to directly obtain normal CAN traces through the OBD-II port in the vehicle, we chose to collect normal CAN traces through the gateway as shown in Fig. 16(a). After identifying the CAN_High and CAN_Low wires responsible for PT-CAN on the gateway based on BMW's circuit diagram, we connect two wires to the P-CAN device. For Renault Koleos, the absence of a gateway allows us to directly obtain normal CAN traces from the OBD-II Port depicted in Fig. 16(b). The OBD-II port is usually positioned beneath the driver's seat. We connect two wires from the P-CAN device to the corresponding pins of CAN_H and CAN_L on the

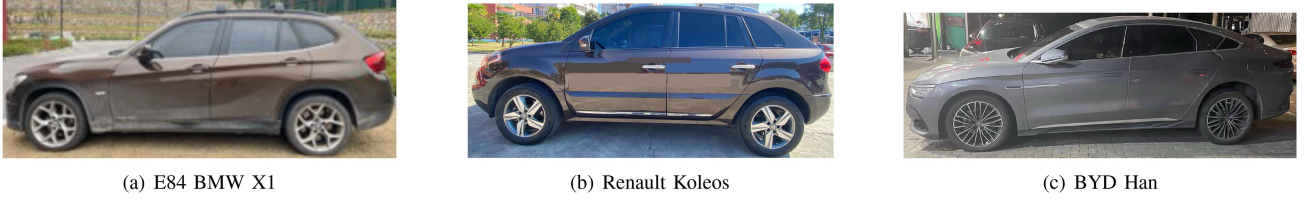


Fig. 14. Test vehicles for evaluation.

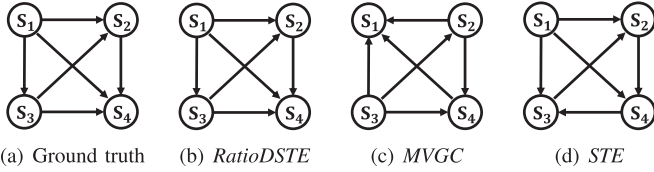
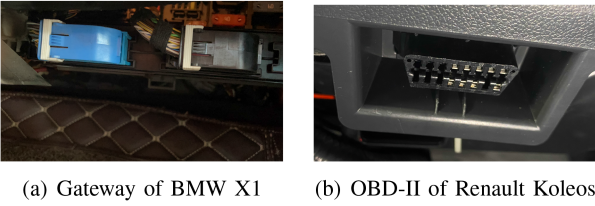
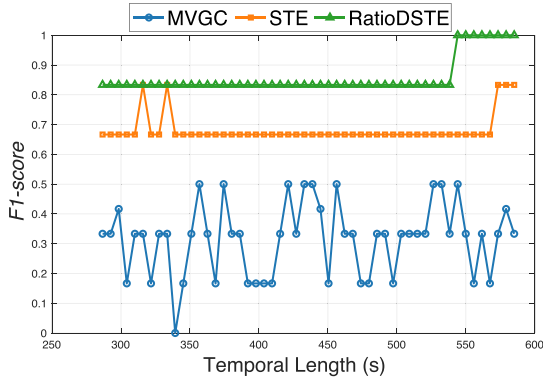
Fig. 15. Summary causal graph generated by *RatioDSTE*-based causal discovery and mainstream methods.

Fig. 16. Data Interface for two ICEVs.

Fig. 17. *F1-score* comparison with different temporal lengths.

OBD-II port. While the volunteer drive the vehicle freely on normal roads for a period of half an hour, the P-CAN device stored tens of thousands of data points with millisecond-level time intervals in the form of 64-bit bitstreams with timestamps.

Semantic Set Extraction: With a normal CAN trace, we need to acquire physical signals transmitted within the data field. This step essentially involves signal filtering, which eliminates signals that lack physical semantics in Fig. 10(b) and (c). Both single-byte and multi-byte encoded physical signals are extracted. Then, physical signals with the same semantics are clustered to form several semantic sets with criteria governed by (1) with $\theta_0 = 0.95$.

The outcome concerning BMW X1 is illustrated in Fig. 18, where the sub-caption represents the ID and boundary for each signal. For instance, as shown in Fig. 18, we use notation $00AA^{(16-23)}$ for single-byte encoding where $00AA$ denotes the CAN ID and its superscript denotes the bit boundary. For multi-byte encoding, we use notation $00BA^{(32-38)*}_{(40-47)}$ where $(\cdot)^*$ denotes the high-order byte.

Fig. 18 also visualizes the semantic sets, where signal $00AA^{(56-63)}$ and signal $0335^{(16-23)}$ belong to one semantic set, signal $00BA^{(32-38)*}_{(40-47)}$ and signal $00CE^{(24-26)*}_{(32-39)}$ belong to another.

The result for Renault Koleos is depicted in Fig. 20. The semantic set division by the same parameter $\theta_0 = 0.95$ is also partially illustrated.

The complete results of signal boundary determination of extracted physical signals for two ICEVs are shown in Table II.

After grouping physical signals into semantic sets, we select the pivot signal for each semantic set based on the time scale calculated by (1). In summary, we significantly abbreviate the search scope of physical signals. For instance, we eliminate the number of candidates from 82 signals to 30 physical signals and finally to 14 semantic sets (pivot signals) in BMW X1, reducing the time required for DSTE calculation.

TECG Generation: Based on the above signal processing results, we take pivot signals representing semantic sets. We calculate the *DSTE* by (3) and further *RatioDSTE* by (4). With $k = 6$ and $t' = 20$, the selected calculation of pairs pivot signals of *RatioDSTE* for BMW X1 is illustrated in Fig. 19. Then, the direction determined in selected results is in the sequence of: $s_2 \rightarrow s_3$, $s_2 \rightarrow s_4$, $s_4 \rightarrow s_3$, $s_9 \rightarrow s_3$ and there is no edge between s_2 , s_9 . We assign value 1 on proper position of matrix $\mathbf{A} \in \mathbb{R}^{14 \times 14}$: $\mathbf{A}(2, 3) = 1$, $\mathbf{A}(2, 4) = 1$, $\mathbf{A}(4, 3) = 1$, $\mathbf{A}(9, 3) = 1$. In Fig. 13, the time delay unit at which pivot signal s_2 has the maximum impact on s_3 , s_4 , and s_7 are different. We assign value $\tau_{\max}$ on the proper position of matrix $\mathbf{E} \in \mathbb{R}^{14 \times 14}$: $\mathbf{E}(2, 3) = 81$, $\mathbf{E}(2, 4) = 52$, $\mathbf{E}(2, 7) = 112$. By iterating through all pairs of pivot signals, the TECG $\mathbf{G}$ is derived with exactly one directed edge between every two nodes. The result for Renault Koleos is partially depicted in Fig. 21. The sequence-directed edges in visualized example between these pivot signals are $s_3 \rightarrow s_1$, $s_1 \rightarrow s_6$, $s_3 \rightarrow s_6$, $s_3 \rightarrow s_8$, $s_8 \rightarrow s_6$ and we fill the $\tau_{\max}$ into the corresponding position of matrix $\mathbf{E}$.

SubGraph Matching: We perform node mapping between the SECG $\mathbf{G}_q$ and the TECG $\mathbf{G}$ based on structure feature defined by matrix $\mathbf{A}$ and edge feature defined by matrix $\mathbf{E}$. We select the SECG depicted in Fig. 5. The visual mapping results for two ICEVs are shown in Fig. 24(a) and (b) where $\mathbf{G}_q$ is on the upper

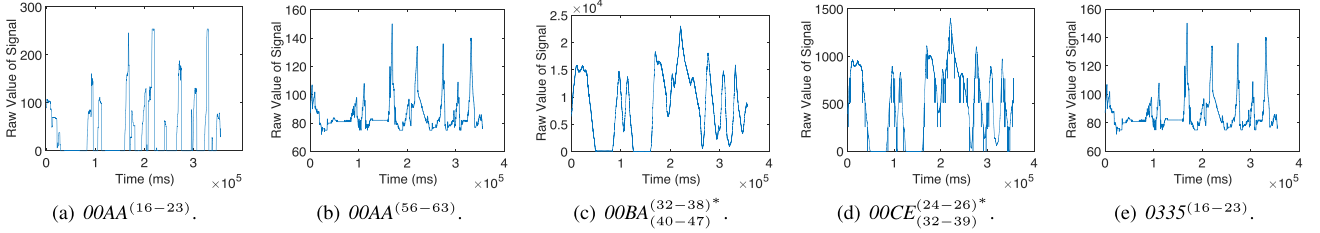


Fig. 18. Visualization of five extracted physical signals clustering into three semantic sets in BMW X1, where physical signal (b), (e) form one semantic set while (c), (d) form another, leaving physical signal (a) alone.

TABLE II
ALL EXTRACTED PHYSICAL SIGNALS BY BFR-MAT BASED ALGORITHM

Physical Signal	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8	f_9	f_{10}
E84 BMW X1	$ID_1^{(16-23)}$	$ID_1^{(32-37)}$	$ID_2^{(24-31)}$	$ID_2^{(40-46)}$	$ID_2^{(56-63)}$	$ID_3^{(40-47)*}_{(48-55)}$	$ID_4^{(16-23)}$	$ID_5^{(48-53)}$	$ID_6^{(32-38)*}_{(40-47)}$	$ID_6^{(8-15)}$
Renault Koleos	$ID_1^{(0-6)}$	$ID_1^{(16-23)}$	$ID_1^{(24-30)}$	$ID_1^{(40-46)}$	$ID_2^{(8-10)*}_{(16-23)}$	$ID_3^{(0-3)*}_{(8-15)}$	$ID_3^{(32-39)}$	$ID_4^{(0-6)}$	$ID_4^{(16-22)}$	$ID_4^{(24-26)*}_{(32-39)}$
BYD Han	$ID_1^{(0-7)}$	$ID_2^{(0-7)}$	$ID_3^{(0-7)}$	$ID_3^{(24-26)*}_{(32-39)}$	$ID_3^{(40-42)*}_{(48-55)}$	$ID_3^{(8-10)*}_{(16-23)}$	$ID_4^{(0-6)}$	$ID_5^{(0-7)}$	$ID_5^{(28-31)}$	$ID_5^{(32-38)}$
	f_{11}	f_{12}	f_{13}	f_{14}	f_{15}	f_{16}	f_{17}	f_{18}	f_{19}	f_{20}
E84 BMW X1	$ID_7^{(24-26)*}_{(32-39)}$	$ID_7^{(40-42)*}_{(48-55)}$	$ID_7^{(8-10)*}_{(16-23)}$	$ID_8^{(16-22)}$	$ID_9^{(48-52)}$	$ID_{10}^{(8-15)}$	$ID_{11}^{(16-20)}$	$ID_{12}^{(49-54)}$	$ID_{13}^{(0-4)}$	$ID_{13}^{(16-20)}$
Renault Koleos	$ID_4^{(40-46)}$	$ID_5^{(32-39)}$	$ID_5^{(40-44)}$	$ID_5^{(56-63)}$	$ID_6^{(16-22)}$	$ID_7^{(32-36)*}_{(40-47)}$	$ID_8^{(0-5)*}_{(8-15)}$	$ID_8^{(16-21)*}_{(24-31)}$	$ID_8^{(32-36)*}_{(40-47)}$	$ID_9^{(0-5)*}_{(8-15)}$
BYD Han	$ID_5^{(40-47)}$	$ID_5^{(8-10)*}_{(16-23)}$	$ID_6^{(56-63)}$	$ID_7^{(24-31)}$	$ID_8^{(24-29)}$	$ID_8^{(8-15)}$	$ID_9^{(16-22)}$	$ID_9^{(56-59)}$	$ID_{10}^{(43-47)}$	$ID_{11}^{(0-5)}$
	f_{21}	f_{22}	f_{23}	f_{24}	f_{25}	f_{26}	f_{27}	f_{28}	f_{29}	f_{30}
E84 BMW X1	$ID_{13}^{(24-28)}$	$ID_{13}^{(8-12)}$	$ID_{14}^{(16-23)}$	$ID_{15}^{(4-7)}$	$ID_{16}^{(16-23)}$	$ID_{16}^{(24-31)}$	$ID_{16}^{(32-39)}$	$ID_{16}^{(48-55)}$	$ID_{16}^{(56-61)}$	$ID_{17}^{(0-7)}$
Renault Koleos	$ID_9^{(16-21)*}_{(24-31)}$	$ID_{10}^{(0-4)*}_{(8-15)}$	$ID_{11}^{(0-4)*}_{(8-15)}$	$ID_{12}^{(0-4)*}_{(8-15)}$	-	-	-	-	-	-
BYD Han	$ID_{11}^{(8-12)}$	$ID_{12}^{(56-63)}$	$ID_{13}^{(0-7)}$	$ID_{14}^{(56-59)}$	$ID_{15}^{(2-6)}$	-	-	-	-	-

- [1] $ID_n^{(a-b)}$: Single-byte encoding where ID_n denotes anonymized CAN IDs (various between different vehicles) and $(a-b)$ denotes signal boundaries within the payload.
 [2] $ID_n^{(a-b)*}$: Multi-byte encoding where $(\cdot)^*$ denotes the high-order byte.

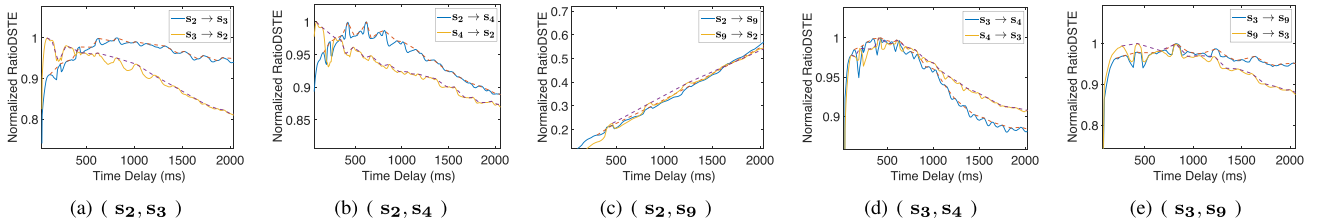


Fig. 19. Normalized *RatioDSTE* between different pivot signals for BMW X1, based on which the causal direction between these pivot signals are determined: $s_2 \rightarrow s_3$, $s_2 \rightarrow s_4$, $s_4 \rightarrow s_3$, $s_9 \rightarrow s_3$ and there is no edge between s_2, s_9 .

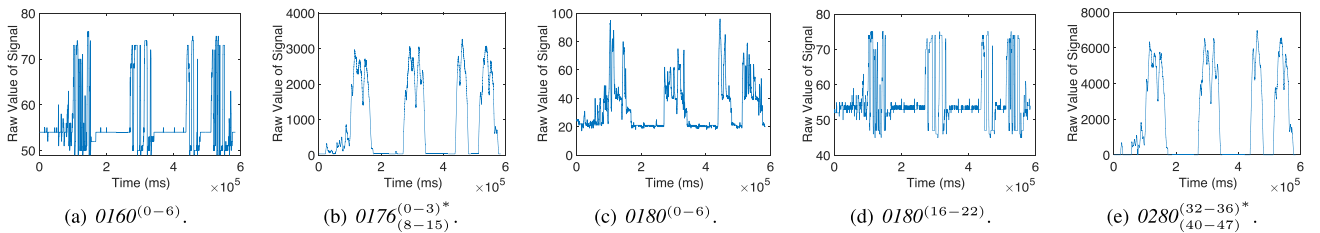


Fig. 20. Visualization of five extracted physical signals clustering into three semantic sets in Renault Koleos, where physical signal (a), (d) form one semantic set while (b), (e) form another, leaving physical signal (c) alone.

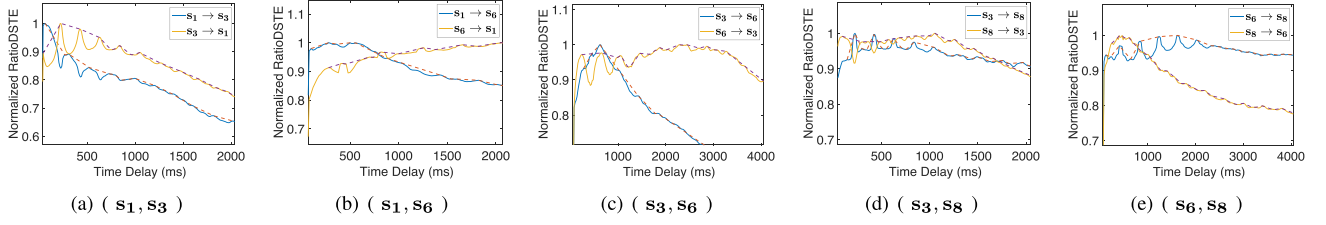


Fig. 21. Normalized *RatioDSTE* between different pivot signals for Renault Koleos, based on which the causal direction between these pivot signals are determined: $s_3 \rightarrow s_1$, $s_1 \rightarrow s_6$, $s_3 \rightarrow s_6$, $s_3 \rightarrow s_8$, $s_8 \rightarrow s_6$.

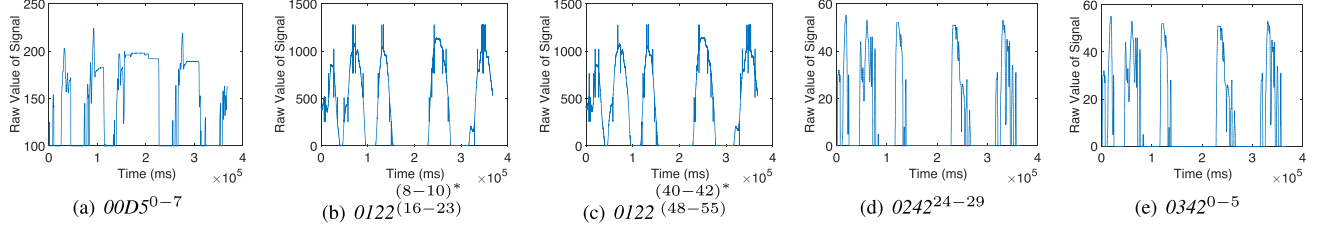


Fig. 22. Visualization of five extracted physical signals clustering into three semantic sets in BYD Han, where physical signal (b), (c) form one semantic set while (d), (e) form another, leaving physical signal (a) alone.

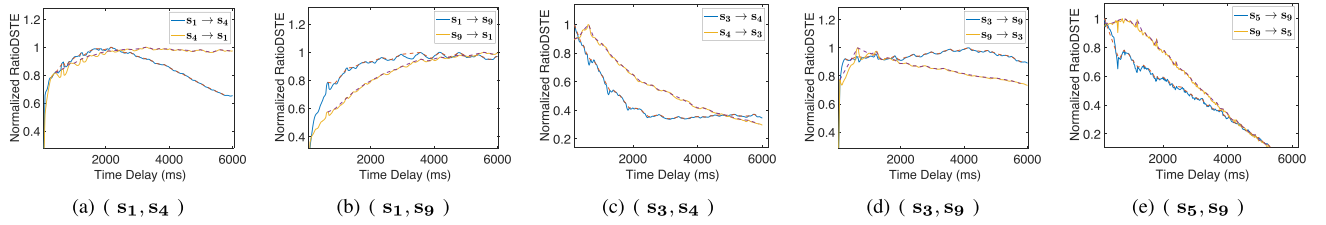


Fig. 23. Normalized *RatioDSTE* between different pivot signals for BYD Han, based on which the causal direction between these pivot signals are determined: $s_1 \rightarrow s_4$, $s_4 \rightarrow s_3$, $s_9 \rightarrow s_3$, $s_9 \rightarrow s_5$ and there is no edge between s_1 and s_9 .

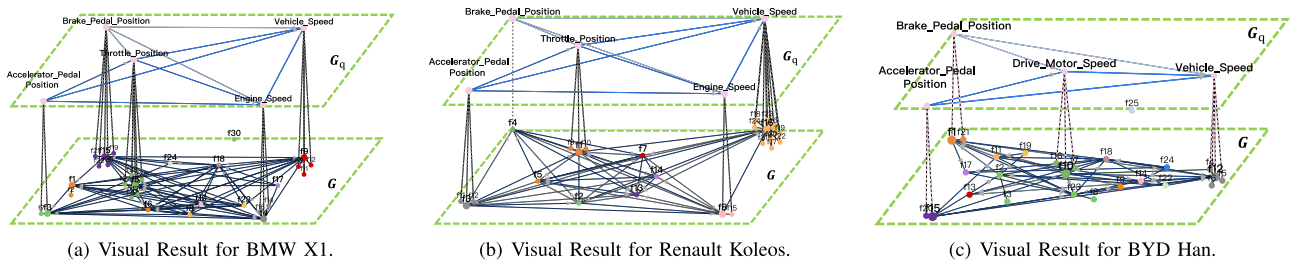


Fig. 24. Overview of reverse engineering results for three automobiles, where SECG is on the upper plane with TEGC beneath.

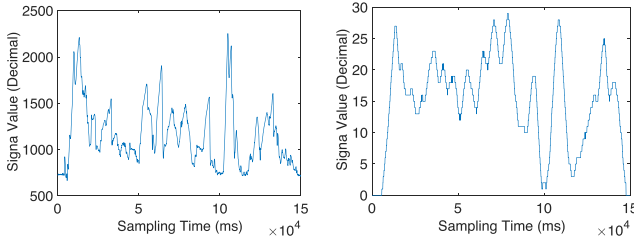
plane with G beneath. In the lower plane, the pivot signals are centered in semantic sets, and the edge between pivot signals is generated with matrix A . Table III displays the full results of semantic reverse engineering for two ICEVs.

Performance of PicaCAN : To verify the correctness of the final results, we collect normal CAN data and diagnostic CAN data simultaneously by connecting the P-CAN device and OBDmeter to PT-CAN during a new driving cycle. The latter is

an STM32 board used to send diagnostic requests to the ECU. We use the diagnostic tool to query the vehicle for four physical signals in our SECG G_q , excluding the brake pedal position for its absence in OBD-II PID. We verify the brake pedal position signal by manual reverse engineering. Four pairs of normal and diagnostic CAN match and we selectively show the engine speed and speed expressed by diagnostic CAN frame and normal CAN frame in BMW X1 in Fig. 25.

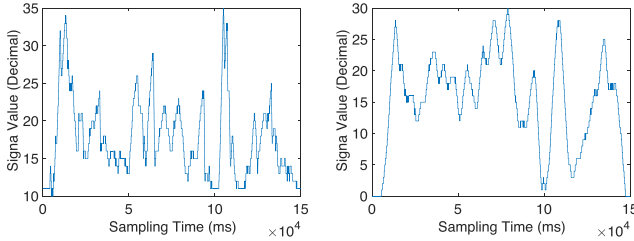
TABLE III
DECODED SEMANTIC INFORMATION OF THE CORRESPONDING PHYSICAL SIGNALS FOR THREE REAL VEHICLES

Vehicle	Accelerator Pedal Position v	f	Brake Pedal Position v	f	Throttle Position v	f	Engine Speed v	f	Vehicle Speed v	f
E84 BMW X1	v ₂	f ₃ , f ₇	v ₉	f ₁₅ , f ₁₉ f ₂₀ , f ₂₁ , f ₂₂	v ₄	f ₅ , f ₂₅ , f ₂₆ f ₂₇ , f ₂₈ , f ₂₉	v ₃	f ₄ , f ₁₄ , f ₁₆	v ₇	f ₉ , f ₁₁ f ₁₂ , f ₁₃
Renault Koleos	v ₃	f ₃ f ₁₁ , f ₁₂	v ₄	f ₄	v ₁	f ₁ f ₉ , f ₁₀	v ₈	f ₈ , f ₁₅	v ₆	f ₆ f ₁₆ ~ f ₂₄
Vehicle	Accelerator Pedal Position v	f	Brake Pedal Position v	f	Drive Motor Speed v	f	Vehicle Speed v	f	-	-
BYD Han	v ₉	f ₁₅ , f ₂₀	v ₁	f ₁ , f ₂₁	v ₄	f ₇ , f ₁₀ , f ₁₆	v ₃	f ₄ , f ₅ f ₆ , f ₁₂	-	-



(a) Engine Speed in Diagnostic CAN

(b) Speed in Diagnostic CAN



(c) Engine Speed in Normal CAN

(d) Speed in Normal CAN

Fig. 25. Reverse engineered signals validated by OBD-II PID.

C. PicaCAN on HEV Under EV Mode

In this subsection, we evaluate PicaCAN on BYD Han under its EV mode shown in Fig. 14(c). Due to the absence of a gateway, we can directly collect normal CAN data through the OBD-II port. Same as before, we conduct the following analysis in offline mode.

The extracted physical signals are selectively shown in Fig. 22, where 5 signals are divided into three semantic sets according to their maximum Pearson coefficient. To address the special case represented in the SECG shown in Fig. 7, we further divide the semantic set containing multiple physical signals to be included in subsequent computations. Then, the calculation result of *RatioDSTE* between semantics sets for BYD Han is selectively displayed in Fig. 23. The sequence directed edges in visualized example between these semantic sets are $s_1 \rightarrow s_4$, $s_4 \rightarrow s_3$, $s_9 \rightarrow s_3$, $s_9 \rightarrow s_5$ and there is no edge between s_1 and s_9 . We fill the $\tau_{\max}$ into the corresponding position of matrix **E**. The visual mapping results for BYD Han are illustrated in Fig. 24(c) and the ultimate reverse engineering findings are given in Table III. Notice that the SECG selected for HEV under EV mode is shown in Fig. 7.

TABLE IV
NUMBER OF EXTRACTED SIGNALS & SEMANTIC TYPE

Experimental Vehicle	E84 BMW X1	Renault Koleos	BYD Han
Continuous Signal	126	60	179
Physical Signal	30	24	25
Decoded Physical Signal	20	19	11
Decoded Semantic Type	5	5	4

D. Summary of Experimental Results

Here, we provide a summary of the experimental results in Tables II–IV.

Specifically, Table II delivers the encoding specifications with anonymized CAN IDs of all physical signals extracted in the experiment. Table III conveys the semantic information of the corresponding physical signal. Furthermore, Table IV provides an overview of the number of signals. We progressively narrow down from hundreds of signals and eventually reverse engineer the semantics of a dozen physical signals (five semantic types for ICEV and four for HEV under EV mode).

To explain why we can extract physical signals representing all targeted semantics in SECG from three vehicles, we analyze the accuracy here. Typically, the accuracy of PicaCAN depends on the accuracy of each individual phase. Among them, *Physical Signal Extraction* and *Subgraph Matching* are rule-based methods while only *TECG Generation* follows a data-driven approach. The specific factors affecting the accuracy of each phase are detailed below:

- *Physical Signal Extraction*: Since physical signals' distinctive features are encoded in the bit-flip-rate-based algorithm and signal labeling, we can extract all the physical signals embedding the semantics on the SECG from the CAN trace. To extract the targeted physical signal, the key is to ensure that its bits undergo flipping during the data collection period and a certain signal magnitude of change occurs compared to Status. Our experiments target the physical signal strongly related to the powertrain system. This implies that these physical signals continuously change with the vehicle's motion. During data collection, the powertrain-related physical signals exhibited significant variations during driving, ensuring that all required

physical signals could be effectively extracted in the experiment.

- *TECG Generation*: The accuracy of TECG Generation is determined by that of *RatioDSTE*, a form of data-driven causal discovery. Grounded in transfer entropy, the accuracy of *RatioDSTE*-based causal discovery relies on the quality of collected data [41]. It means that the data must inherently reflect the causal relationships among the signals. Our experiment aims to capture physical signals that semantically describe how power is transmitted to the wheels when the accelerator is pressed. Therefore, the causal relationships within the powertrain system are inherently embedded in the CAN trace during vehicle operation. Meanwhile, the quantity of data is also important in revealing causal relationships accurately, which is illustrated in Fig. 17.
- *Subgraph Matching*: As we use the exhaustive search method, Subgraph Matching can generate an accurate result as long as SECG within TECG is unique. In our experiment, the interaction pattern of the generated SECG is unique in the corresponding vehicle's PT-CAN. Specifically, since the number of physical semantics within PT-CAN is limited (in our experiments, the semantic sets obtained through clustering physical signals are all fewer than 20), we apply brute force search to find the isomorphic subgraph to SECG in TECG. Since the node and edge weights of SECG and TECG follow the same criteria, SECG can certainly be identified. Note that if SECG is not unique in TECG, Subgraph Matching can produce incorrect results. The potential solution is to make the SECG more complex according to the two *criteria* outlined in Section III, ensuring that the newly constructed graph is unique.

VII. DISCUSSION

A. Potential Applications of PicaCAN

The primary application of PicaCAN is to provide security researchers or (white hat) hackers from third parties with a more efficient solution to reverse engineer the physical semantics of safety-critical signals within vehicle powertrain systems. These safety-critical signals are essential prerequisites for defense and attack methodologies such as state-aware intrusion detection systems [42] and stealthy attacks targeting GPS or engine speed [7], [43]. Since PicaCAN enables extracting physical semantics without the need for conducting the interaction (such as PIDs, challenge actions, or signals), the time and human effort required for reverse engineering will be greatly reduced. Therefore, the defense or attack methodologies developed will facilitate more rapid adaptation to various vehicle models. Another possible use-case we foresee for PicaCAN is providing powertrain sensory data, which serves as an irreplaceable component in the modern automotive domain. These sensory signals establish a solid foundation for both In-Vehicle and Vehicle-to-Everything (V2X) Communication, supporting richly advanced functions and services including fleet management, traffic analysis, cloud services, and Advanced Driver Assistance (ADAS) systems [44], [45].

PicaCAN could also be extended to other protocols. Note that it can be directly adapted to these protocols that use the CAN protocol as the underlying transport protocol in different scenarios. For example, PicaCAN can be a more efficient framework for reverse engineering diagnostic protocols like UDS in an automotive system [46], proprietary devices/messages related to NMEA 2000 in shipboard systems [47], [48], and vendor-specific peer messages within DeviceNet [49] in industrial control systems. For protocols unrelated to CAN protocol, the part design of PicaCAN, such as matching the causal graph generated from the data and the prior knowledge, could be used if the targeted protocol has a well-defined payload boundary and the extract signals interact with delayed impact.

B. Limitation and Future Improvement

- *Limitation 1*: PicaCAN still requires some manual intervention, especially in the phase of SECG construction. Users need to express the targeted physical signals in the form of SECG based on knowledge of time precedence between these signals.
- *Future Improvement for Limitation 1*: One potential solution to this issue is to use knowledge graph techniques. Naturally modeling entities and their relationships as a graph structure, knowledge graphs are well-suited for establishing relationships between signals within vehicles, capturing both hierarchy and interconnectedness [5]. The challenge lies in extracting the relationships between physical signals within vehicles from the dataset, as no explicit data is available to provide this information. Instead, one must characterize this by looking for subtle clues in semi-structured data sources (e.g., vehicle log files and event records) and unstructured data (e.g., technical manuals, and maintenance logs).
- *Limitation 2*: PicaCAN is specifically designed for extracting physical signals within the powertrain system. Thus, the number of physical signals that can be decoded largely depends on the SECG generated, i.e., PicaCAN can only extract physical signals having delay-impacted interactions with each other in vehicle dynamics.
- *Future Improvement for Limitation 2*: To reverse engineer more physical signals within CAN trace, more knowledge of the vehicle operation mechanism is essential to construct a comprehensive SECG. However, the specific knowledge of the subsystem mechanism in the vehicle is vendor-specific compared to the powertrain system. Meanwhile, users must identify when the subsystem is activated, unlike the logic in the powertrain which operates continuously. To reverse engineer status signals, one possible solution is to combine physically induced causality with *reactive parametric signal temporal logic* (rPSTL) [50]. Both physical and status signals are described using signal temporal logic ϕ and connected via temporal operators including $\Diamond_{[a,b]}\phi$ (read “eventually ϕ ”) and $\Box_{[a,b]}\phi$ (read “always ϕ ”), where $[a, b]$ refers to the time interval. For instance, the fact that the door locks automatically engage when the driver's speed exceeds a certain value could be expressed

by rPSTL as

$$\Diamond_{[0,\tau_1]} (\Box_{[0,\tau_2]} (v > v_0) \rightarrow \Diamond_{[\tau_3,\tau_4]} (x_{\text{locked}} = 1)), \quad (9)$$

which means that: If there is a time t in $[0, \tau_1]$ such that the vehicle speed exceeds the threshold v_0 for the next τ_2 seconds, then the door locked signal is activated in at least τ_3 and at most τ_4 seconds. The challenge lies in the determination of the coefficients $(\tau_1, \tau_2, \tau_3, \tau_4, v_0)$, which can be vender-specific.

VIII. RELATED WORK

CAN Message Tokenization: Over the past five years, researchers have accomplished much in the tokenization of the CAN protocol, the objective of which is to identify signal boundaries and encoding endianness within the CAN message's payload. Markovitz first tackled this by adopting a brute force method to search every potential token boundary combination [19]. The extremely high computational expenditure prevents its widespread adoption. Verma further developed combinatorial optimization by ACTT [10], reducing complexity from $\mathcal{O}(n^2)$ to $\mathcal{O}(n \log n)$. The limitation of this study is the intrusiveness involved in injecting diagnostic parameter IDs and misperception towards the endianness of CAN messages. Another prevalent approach called READ uses bit flip rate (BFR), which was first proposed by Marchetti and Stabili [26]. Reducing the complexity to $\mathcal{O}(n)$, READ reaches the correct percentage over 90% in extracting physical signals. Further improved by LibreCAN [11], Pesé established a percentage threshold to validate the reduction in BFR and defined two metrics for the evaluation of tokenization.

CAN Message Semantic Translation: Semantic translation focuses on reverse engineering the physical semantics of signals transmitted in the CAN bus. External information-based methods originated from Huybrechts [9] who incorporated the internal diagnostic protocol (mostly OBD-II) to offer extra features. LibreCAN [11] appeared to be the epitome of this category, offering a complete workflow to integrate information from PIDs, Inertial Measurement Units, and GPS data to search the correspondence physical signals within CAN frame.

AutoCAN [12] reduced the need for external information, utilizing only external sensors and devices to identify certain signals through Pearson correlation analysis. Subsequently, the identified signals undergo further mathematical processing using formulas searching for more physical signals. Except for sensor information, CANMatch [1] fully leveraged non-sensor information, previously reverse-engineered DBC files, as prior knowledge. It is established upon the assumption that ECU models installed by manufacturers on any two vehicles not only send frames with identical vehicle functionalities but also send frames with the same ID. By breaking down a portion of the signals, it then employs correlation analysis to identify new signals. This method is limited by acquiring a large number of DBC file inputs. The learning-based method was first adopted in semantic reverse engineering by Jaynes who used supervised learning with features including inherent characteristics of the CAN messages (identifier, payload, frequency) to cluster semantic meanings [51]. Failing to translate individual signals,

Jaynes' work was modified by Buscemi who designed a Critical Signals Identifier to compensate for this shortcoming [52]. The supervised learning-based method is significantly constrained by training datasets and has not been validated in real-life datasets. Ezeobi [53] and Young [54] accomplished the task of coarse-grained clustering using unsupervised methods, which yet have not been delved into the physical semantics of individual signals.

IX. CONCLUSION

In this paper, we propose PicaCAN an automatic tool to reverse engineer the semantics of CAN messages on PT-CAN. Merely based on CAN traffic, PicaCAN reverses engineer the encoding specification of powertrain-related signals on PT-CAN, offering researchers an eye vehicle's powertrain system. To the best of our knowledge, PicaCAN is the first graph-based tool that combines cyber and physical knowledge to reverse engineer PT-CAN. We design an information-theory-based causal discovery method, *RaitoDSTE*, to capture the impact delay between various powertrain-related signals and utilize it to define the physically induced causalities from a pure data perspective. We experimentally verified PicaCAN on PT-CAN of three real vehicles of two energy types and successfully reversed out their different encoding specifications of powertrain-related signals.

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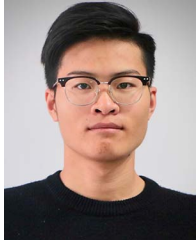
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